The Geometric Atom: Atomic Structure as a Packing Problem*

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We present a discrete geometric framework for the hydrogen atom based on finite information capacity. The hydrogen state space is modeled as a paraboloid lattice dual to the SO(4,2) conformal algebra. This single-particle model exhibits a fundamental algebra-geometry duality: **(i) Algebraic exactness:** [**PANEL-VERIFIED**] Transition operators $T_\pm, L_\pm$ reproduce the Rydberg quantum numbers from operator eigenvalues, bit-exactly: the $SU(1,1)$ element $N = -2[T_+, T_-]$ returns n on the interior shells $n < n_{\max}$ (not on the truncated top shell), and L^2 returns $\ell(\ell+1)$ on every shell; on every shell, including the top one, the $SO(4)$ Casimir $\mathbf{L}^2 + \mathbf{A}^2 = n^2 - 1$ ($\mathbf{A}$ the energy-normalized Runge–Lenz vector) is exact on the labels. The energies $E_n = -1/(2n^2)$ are an imposed map from n (Fock's energy shell $p_0^2 = -2E$), not an output of the operators. **(ii) Geometric artifacts:** [**MEASURED**] A node-amplitude proxy for the s/p gap is nonzero at finite lattice cutoff (37% at $n_{\max} = 5$, oscillating through 15.6% at 8 and 1.7% at 10 to 0.39% at 30 on the binary production lattice; see the §III QA note). [**INTERNAL THEOREM**] Its size tracks the residue of $n_{\max}$ modulo 3 — exactly closed form on $n_{\max} \equiv 0$, and generically larger off it (the earlier $n_{\max}/(\sqrt{3}\,\pi)$ asymptotic is [retracted 2026-09-04: sp-ratio-false-precision], read off three sampled cutoffs) — so the quoted endpoint sits on a favourable branch and the sequence is not evidence of convergence (2026-09-03; the earlier wording, "the graph Laplacian lifts s/p degeneracy ... through differential weighted connectivity", both contradicted the clause below and named the weighted construction's mechanism for binary-lattice numbers, where every edge weight is 1). The cutoff sweep from $n_{\max} = 5$ to 30 shows this is a discretization artifact—a residue-controlled truncation effect, *not* spectral aliasing [retracted 2026-09-04: sp-splitting-aliasing-mechanism] (that reading was withdrawn 2026-09-03: the l-blocks are disconnected and carry no degeneracy to alias)—that oscillates and reads 0.39% at $n_{\max} = 30$ on the $n_{\max} \equiv 0 \pmod 3$ branch (1.65% and 2.58% at 28 and 29): a proxy value, not evidence of recovering the exact Coulomb degeneracy [retracted 2026-09-04: sp-splitting-aliasing-mechanism], which belongs to the continuum operator (the graph's l-blocks are disconnected, so the graph has no s/p degeneracy of its own). **(iii) Geometric phase structure:** [**SYMBOLIC PROOF**] The Berry phase vanishes identically for the real-valued transition operators; the edge-weight log-holonomy $\Theta(n) = -2\ln((n+1)/n) \sim n^{-1}$ measures the curvature of the weight function across the lattice. **Crucially, this model describes only the electron manifold**—electromagnetic interactions and the fine structure constant α cannot emerge from a single-particle lattice and require coupling to photon degrees of freedom, which we address in a companion paper.

I. INTRODUCTION: THE TEXTURE OF THE VACUUM

Physics traditionally assumes spacetime is a smooth continuum. Quantum mechanics inherits this assumption, treating the Hilbert space as infinite-dimensional with continuous operators. Yet this framework produces persistent conceptual difficulties—wave-particle duality, nonlocality, and the measurement problem—that resist resolution within the continuum paradigm.

We propose an alternative foundation: *physical systems encode finite information.* A quantum state labeled by quantum numbers (n, l, m) cannot store infinite positional precision. If information density is finite, then the state space itself must be *discrete*—a lattice, not a continuum. The hydrogen state space is then indexed by a **graph**: nodes are the quantum-number labels of the basis states, edges connect labels related by the ladder operators $T_\pm, L_\pm$, and those operators act as sparse matrices on the labels.

This discrete structure possesses a natural geometry. The hydrogen atom's $SO(4)$ symmetry [2], extended to the $SO(4,2)$ dynamical group [1], has a discrete dual: the **paraboloid lattice**, where quantum numbers (n, l, m) map to coordinates on a curved 3D surface. The radius grows parabolically as $r \sim n^2$, and the depth encodes energy: $z = -1/n^2$.

This lattice exhibits a profound duality between *algebra* and *geometry*:

- **The Algebraic Skeleton:** Transition operators $T_\pm$ (radial) and $L_\pm$ (angular) generate the quantum numbers. Their eigenvalues reproduce n exactly on the interior shells and $\ell(\ell+1)$ on every shell (on every shell, including the top one, the $SO(4)$ Casimir $n^2 - 1$ is exact on the labels); the formula $E_n = -1/(2n^2)$ is then an imposed map from n through Fock's energy shell, not an output of the operators.
- **The Geometric Flesh:** The lattice *topology*—the pattern of connections—produces finite-truncation

* Reconstructing Atomic Structure from the Principle of Finite Capacity

artifacts, not physical corrections: the structure of the graph Laplacian is intrinsic to its disconnected two-dimensional grid blocks. High angular momentum states occupy higher-degree nodes at finite truncation (read before 2026-09-03 as differential weighted connectivity [retracted 2026-09-04: sp-splitting-aliasing-mechanism] lifting the s/p degeneracy; withdrawn — the l-blocks are disconnected components). The edge-weight log-holonomy provides a measure of lattice curvature that decays as n^{-1}.

In this paper, we present computational evidence for this dual framework. Section II demonstrates exactness of the quantum numbers on interior shells. Section III analyzes the s/p node-amplitude proxy, showing it is a discretization artifact whose size tracks the residue of $n_{\max}$ modulo 3 — not a graph degeneracy (the ℓ-blocks are disconnected) and not a continuum-convergence signal. Section IV analyzes the geometric phase structure, showing that the Berry phase vanishes while the log-holonomy decays as n^{-1}. Section V discusses the algebra-geometry duality and its connection to wave-particle complementarity. Section VI summarizes our findings.

II. THE ALGEBRAIC SKELETON: EXACT QUANTUM NUMBERS

The paraboloid lattice is generated by two sets of ladder operators forming the hydrogen atom's dynamical algebra $SU(2) \otimes SU(1,1)$:

$$\text{Angular:} \quad L_z, L_\pm \quad \text{(standard angular momentum)}, \tag{1}$$

$$\text{Radial:} \quad T_0, T_\pm \quad \text{(energy shell transitions)}. \tag{2}$$

These operators act on basis states $|n, l, m\rangle$ with selection rules:

$$L_\pm |n, l, m\rangle \propto |n, l, m \pm 1\rangle, \tag{3}$$

$$T_\pm |n, l, m\rangle \propto |n \pm 1, l, m\rangle. \tag{4}$$

The operator weights are determined by the Biedenharn-Louck calculus [3] for the $SU(2)$ Clebsch-Gordan coefficients and by the noncompact $SU(1,1)$ ladder structure [1].

A. Quantum-Number Reproduction

The critical test of this algebra is spectroscopic: do the operators reproduce the quantum numbers that label the Rydberg formula? **[PANEL-VERIFIED]** They do, on interior shells: the operators return n ($n < n_{\max}$) and $\ell(\ell+1)$ bit-exactly, and the Rydberg energies

$$E_n = -\frac{1}{2n^2} \quad \text{(Hartree atomic units)}, \tag{5}$$

are then the imposed map from n given by Fock's energy shell $p_0^2 = -2E$ (Paper 7), not an eigenvalue output of the lattice. (The matched kinetic scale $\kappa = -1/16$ plays no role here: it matches only the graph Laplacian's extremal eigenvalue, a band edge, to $-\frac{1}{2}$, and its coincidence with the Fock Jacobian factor is an Observation, Paper 7.) No fitting parameters are required; the algebra determines the quantum-number structure uniquely.

This result validates the lattice as a faithful discrete *labelling* of the (truncated) hydrogen Hilbert space. The nodes are not a coarse-graining—they label the quantum states. The edges connect labels related by the ladder operators $T_\pm, L_\pm$; they are not dipole-allowed physical transitions. The quantum-number structure emerges from operator eigenvalues, not from solving differential equations; the energies are the imposed energy-shell map of n. (Verified in `tests/test_paper1_rydberg.py`: the $SU(1,1)$ number operator $N = -2[T_+, T_-]$, assembled purely from the ladder generators, has integer spectrum $\{n\}$ on the *interior* shells $n < n_{\max}$ and the $SU(2)$ Casimir L^2 gives $l(l+1)$, both bit-exact; $E_n = -1/(2n^2)$ is then imposed as a function of n, not output by the algebra. This is the dynamical-symmetry construction, and it has three limits stated here so that it is not over-read: its n-dependence enters through the off-diagonal $T_\pm$ weights rather than through a diagonal operator; it is not exact on the top shell, where the truncated commutator makes N l-dependent and non-positive (the test checks interior states only); and $SU(1,1)$ is noncompact, so its truncation is not a subrepresentation. The exact *compact* discrete operator on these labels is the $SO(4)$ Casimir $\mathbf{L}^2 + \mathbf{A}^2 = n^2 - 1$ ($\mathbf{A}$ the energy-normalized Runge–Lenz vector), diagonal with multiplicity n^2 on every shell including the top one, since each shell is a complete irreducible representation (`tests/test_graph_s3_nonconvergence.py`); it is $-\Delta_{S^3}$, the S^3 Laplace–Beltrami operator up to sign, acting on the label set. The discrete graph Laplacian $L = D - A$ is a separate object and does *not* reproduce this spectrum in any limit: it preserves l, splits into $n_{\max}$ disconnected blocks, and does not converge to Δ_{S^3} (Paper 7 [8]; corrected 2026-09-25 from "reproduces the spectrum only in the continuum limit").)

III. THE GEOMETRIC FLESH: TOPOLOGICAL ARTIFACTS OF FINITE TRUNCATION

While the algebraic operators yield the exact quantum numbers (on interior shells), the *geometric structure* of the lattice—its connectivity pattern—introduces systematic artifacts when the lattice is truncated at finite $n_{\max}$. We demonstrate this by constructing the graph Laplacian and analyzing its symmetry-breaking properties as a function of lattice size.

A. The Graph Laplacian Hamiltonian

In graph theory, the kinetic energy of a particle hopping on a graph's vertices is described by the **graph Laplacian** [5]; here the vertices are quantum-number labels, not positions, so this is a statement about the graph, not about the electron:

$$L = D - A, \tag{6}$$

where A_{ij} is the adjacency matrix (edge weights) and D is the weighted degree matrix[9]. This Laplacian is a discrete analog of $-\nabla^2$ on each l-block (a two-dimensional grid in n and m); it is not a discretization of the S^3 Laplacian, to which it does not converge (Paper 7).

For the paraboloid lattice, we construct A from the transition operators:

$$A = |T_+| + |T_-| + |L_+| + |L_-|, \tag{7}$$

where $|\cdot|$ denotes element-wise absolute value (undirected edges). The Hamiltonian becomes

$$H = \beta L + V = \beta(D - A) + V, \tag{8}$$

where $V_{ii} = -1/n_i^2$ is a diagonal Coulomb node weight and β a scaling factor. **Eq.** (8) **is a general form, not the object measured below and not the production object** (clarified 2026-09-03): the splitting measurements of §III are taken on the bare graph Laplacian $L = D - A$, and the labelled legacy graph model `AtomicSolver(hamiltonian='graph')` builds $H = K_{\text{vac}}L$ with node weights never applied (the solver's default is now the physical diagonal $H = \text{diag}(-Z^2/2n^2)$ on the labels; the splitting numbers below are measured on the bare lattice and are unaffected). Under Eq. (8) with $V_{ii} = -1/n_i^2$ the same twelve cutoffs give 44.7, 2.2, 27.5, 10.6, 31.5, 620.9% and above — a different quantity entirely.

B. Differential Connectivity by Angular Momentum

The crucial observation is that the degree matrix D is *not rotationally invariant*. Computing the weighted degrees for states with $n = 2$:

$$\text{Pole (2s):} \quad D_{(2,0,0)} = 0.854, \tag{9}$$

$$\text{Equator (2p):} \quad D_{(2,1,0)} = 3.416. \tag{10}$$

[PANEL-VERIFIED] States with higher angular momentum l have *more connections*—more available transitions to neighboring states. This higher connectivity translates to higher on-site energy in the Laplacian formulation; it does not couple the l-blocks, which are disconnected components, so it is not a mechanism for the s/p proxy (that reading withdrawn 2026-09-03; §III).

C. The Splitting as Discretization Artifact

We performed exact diagonalization of the bare graph Laplacian $L = D - A$ for lattices with $n_{\max}$ ranging from 5 to 30 (corrected 2026-09-03: this paragraph read "of H (Eq. 8)", which is not the operator measured — see the note there). Using a full dense eigendecomposition, we identified the eigenstates with maximal overlap with $|2,0,0\rangle$ (2s) and $|2,1,0\rangle$ (2p) and computed the relative splitting $\Delta E_{\text{rel}} = |E_{2p} - E_{2s}|/|E_{2s}|$.

At $n_{\max} = 10$ (385 nodes), the *originally reported* splitting was:

$$\lambda_{2s} = 0.0202, \tag{11}$$

$$\lambda_{2p} = 0.0238, \tag{12}$$

$$\Delta E = \lambda_{2p} - \lambda_{2s} = 0.0035, \tag{13}$$

quoted as a 16% relative splitting (arithmetically 17.3%). **[MEASURED]** *These are not the bare Laplacian's values* and are retained only as the historical record: on the operator named above, $L = D - A$ at $n_{\max} = 10$ gives $\lambda_{2s} = 3.1756$, $\lambda_{2p} = 3.1206$ (corrected 2026-09-04 from a transposed 3.2306 — its mirror about λ_{2s}, which is why the ratio was unaffected) and a 1.73% ratio (see the QA correction below, and the twelve-cutoff list in §III). Flagged inline 2026-09-04: the method sentence above was corrected to name $L = D - A$ while this display still carried the legacy figures, so the corrected sentence appeared to vouch for them. However, this degeneracy breaking is *not* an emergent physical force. Systematic convergence analysis reveals it is a **discretization artifact** of finite lattice truncation—the graph-theoretic analog of spectral aliasing [retracted 2026-09-04: sp-splitting-aliasing-mechanism] in finite-element methods, an analogy retired for the s/p proxy on 2026-09-03 (the ℓ-blocks are disconnected, so nothing is aliased).

a. QA correction (2026-06-14). **[MEASURED]** The specific quantitative values in this section—the degrees $D_{2s} = 0.854$, $D_{2p} = 3.416$, the eigenvalues $\lambda_{2s} = 0.0202$, $\lambda_{2p} = 0.0238$, and the 16% splitting quoted at $n_{\max} = 10$—are *not* reproduced by the current production lattice (the $A = |T_+| + |T_-| + |L_+| + |L_-|$ CG-magnitude construction they assume is not implemented in `geovac/`; a faithful rebuild of that construction gives $D_{2s} \approx 1.9$ and $D_{2p} \approx 3.8$). Two lattices must be kept apart (2026-09-03): on the CG-weighted construction the $2s/2p$ splitting does *not* decay—129%, 68%, 84% and 139% at $n_{\max} = 8, 10, 15, 20$—whereas on the *binary* production lattice (`GeometricLattice`, all edge weights 1, the object used by the legacy graph mode `AtomicSolver(hamiltonian='graph')`) it is 37%, 12.7%, 0.65%, 15.6%, 5.1%, 1.7%, 2.7%, 1.7%, 1.1%, 4.1%, 1.8% and 0.39% at $n_{\max} = 5, 6, 7, 8, 9, 10, 12, 15, 18, 20, 25, 30$. Six of those twelve cutoffs are multiples of 3, including the endpoint, and on that residue class the value is exactly $\left(2 - 2\cos\frac{\pi}{n_{\max}-1}\right)/3$ (Eq. (14)); the neighbouring cutoffs 28 and 29 give 1.65% and 2.58%.

What *is* verified (`tests/test_trunk_qa_splitting.py`, `test_trunk_qa_degree_table.py`) is therefore: higher-l states have higher connectivity ($D_{2p} > D_{2s}$) on the CG construction, and on the binary lattice the s/p splitting decays, strongly oscillating, to below 1% by $n_{\max} = 30$. The specific CG-construction figures are retained as illustrative of that construction only. *This caveat applies wherever these figures recur—Table I (the abstract, Appendix B and the convergence list below now print the binary-lattice measurements). (The placeholder figure environments that also restated them were removed 2026-09-01.)*

Conditioning caveat (2026-09-03). The l-blocks of the graph Laplacian are *disconnected* components (no edge changes l), so there is no $2s/2p$ degeneracy in the graph spectrum to lift: the quantity reported here selects, for each of the nodes $|2,0,0\rangle$ and $|2,1,0\rangle$, the eigenvector of maximal amplitude at that node, and that selection is a near-tie (top-two overlaps within 1.2% at $n_{\max} = 30$, with selected eigenvalues 3.0 and 0.011). The percentages below reproduce exactly but are *not robust* to the mode-selection rule; they are a node-amplitude proxy, not a spectral gap (`tests/test_paper1_block_spectrum.py`).

D. Convergence Analysis

The key diagnostic is the behavior of $\Delta E_{\rm rel}$ as $n_{\max} \to \infty$. **[MEASURED]** We swept the lattice cutoff from $n_{\max} = 5$ (55 vertices) through $n_{\max} = 30$ (9,455 vertices) and tracked the 2s/2p splitting at each resolution:

- At $n_{\max} = 5$: $\Delta E_{\rm rel} \approx 37\%$ (the maximum on the grid)
- At $n_{\max} = 6, 7$: 12.7%, 0.65%
- At $n_{\max} = 8$: 15.6% (a secondary peak); by $n_{\max} = 10$ it has fallen to $\approx 1.7\%$
- At $n_{\max} = 15, 18$: 1.7%, 1.1%
- At $n_{\max} = 20$: $\Delta E_{\rm rel} \approx 4.1\%$ (a third excursion)
- At $n_{\max} = 25, 30$: 1.8%, 0.39%

(binary production lattice, full-spectrum maximum-overlap identification — see the conditioning caveat above; `tests/test_trunk_qa_splitting.py` pins the qualitative content (bounded, non-monotone, downward trend, sub-percent at $n_{\max} = 30$), `tests/test_paper1_block_spectrum.py` the conditioning and the CG-construction non-decay; the list printed before 2026-09-03—13%, 16%, 0.3%, 0.005%—did not reproduce at $n_{\max} = 5$, 20 or 30.)

The splitting does not decay monotonically; it *oscillates* at intermediate resolutions. **[INTERNAL THEOREM]** *The oscillation is the residue of $n_{\max}$ modulo 3, and the sequence is closed form* (2026-09-03). The $\ell = 0$ block is the path $P_{n_{\max}}$, whose eigenvalues are $2 - 2\cos(j\pi/n_{\max})$; the mode selected by maximal overlap with the $2s$ node has $j = \text{round}(2n_{\max}/3)$, so

$$\lambda_{2s} = 3 \textit{ exactly} \quad \Longleftrightarrow \quad n_{\max} \equiv 0 \pmod 3, \tag{14}$$

and on that branch the reported ratio is exactly $\big(2 - 2\cos\frac{\pi}{n_{\max}-1}\big)/3$ — 2.7005, 1.6715, 1.1351, 0.8208, 0.6209, 0.4861 and 0.3908% at $n_{\max} = 12, 15, 18, 21, 24, 27, 30$, matching the measured values to every digit printed. *At neighbouring cutoffs* the off-branch value is generically larger — asymptotically by $n_{\max}/(\sqrt{3}\,\pi)$ [retracted 2026-09-04: sp-ratio-false-precision], but far from that at small cutoff: 4.2× and 6.6× at $n_{\max} = 28, 29$ against 30 (1.65%, 2.58% vs 0.39%), where the formula gives 5.5; 20–23× at $118, 119$ against 120 (22.1); 550–552× at $2998, 2999$ against 3000 (551.3). Below $n_{\max} \approx 12$ the ordering can invert (0.65% at 7 against 12.7% at 6). (Two earlier statements of this ratio — "three to six times" and "tracking $n_{\max}/(\sqrt{3}\,\pi)$ to within 1%" [retracted 2026-09-04: sp-ratio-false-precision] — were each read off three sampled cutoffs and stated as theorems; both are withdrawn.) **The endpoint quoted throughout this paper,** 0.39% **at $n_{\max} = 30$, therefore lies on the favourable branch**, a selection this paper did not disclose before 2026-09-03.

The decay is consequently $O(n_{\max}^{-2})$ along $n_{\max} \equiv 0$ and $O(n_{\max}^{-1})$ generically — a proved rate, replacing the earlier statement that none was proven. *The mechanism previously printed here — "spectral aliasing… standing-wave reflections that differentially [retracted 2026-09-04: sp-splitting-aliasing-mechanism] shift eigenvalues in distinct angular momentum sectors" — is withdrawn:* the ℓ-blocks are disconnected components (`tests/test_paper1_block_spectrum.py`), so there is no s/p degeneracy present to alias and no cross-sector reflection to shift it. Both selected eigenvalues approach $3 = 2 - 2\cos(2\pi/3)$ because the target node is vertex 2 of a path in each block. As $n_{\max}$ grows the two proxy eigenvalues coincide (a path-graph coincidence), which does *not* reproduce the continuum $SO(4)$ Coulomb degeneracy (the ℓ-blocks are disconnected, so the graph has no s/p degeneracy of its own).

E. Physical Interpretation

[PANEL-VERIFIED] The 2p state has the higher weighted degree ($D_{2p} > D_{2s}$; the "4×" ratio came from the non-reproducing Table I figures — see the §III QA note). *Read before 2026-09-03 as:* the higher-degree 2p node is more sensitive to the lattice boundary than the 2s state, the finite-$n_{\max}$ truncation acting as an absorbing wall that differentially perturbs high-connectivity nodes [retracted 2026-09-04: sp-splitting-aliasing-mechanism]. *Withdrawn for the s/p proxy (2026-09-03):* its size is set by the residue of $n_{\max}$ modulo 3, not by a boundary effect; the "geometric boundary effect" reading printed here until 2026-09-04 is withdrawn with it, and it is not an emergent centrifugal barrier either.

Crucially, *no explicit* $l(l+1)/r^2$ term appears in the Hamiltonian. The transient splitting arises entirely from the finite extent of the graph, not from any force encoded in the topology. The graph Laplacian does not spontaneously generate the Lamb shift or replace electromagnetic interactions, and its finite-lattice s/p proxy does *not* reproduce the continuum $SO(4)$ Coulomb degeneracy (the ℓ-blocks are disconnected components, so the graph Laplacian has no continuum limit carrying that degeneracy—it does not converge to Δ_{S^3}, Paper 7; the degeneracy is carried exactly by the $SO(4)$ Casimir on the labels, §II). **[INTERNAL THEOREM]** The s/p node-amplitude proxy is a characterisable discretization artifact with a closed-form value on $n_{\max} \equiv 0 \pmod 3$ and a proved rate — $O(n_{\max}^{-2})$ there, $O(n_{\max}^{-1})$ generically (Eq. (14); the earlier "no convergence rate is proven" is superseded). Its magnitude depends on the residue of $n_{\max}$ modulo 3 rather than on proximity to a continuum limit, so it does not function as evidence of convergence; see the §III QA note.

IV. GEOMETRIC PHASE STRUCTURE OF THE LATTICE

The lattice carries a natural notion of parallel transport around closed loops (plaquettes). We examine whether this transport produces a nontrivial geometric phase—a Berry holonomy [4]—and find that the answer is negative for the real-valued transition operators used here. We then characterize the *log-holonomy*, which measures curvature of the edge-weight function, and discuss what would be required for a genuine geometric phase.

A. Flat Connection for Real Operators

A **rectangular plaquette** on the lattice is constructed via radial ($T_\pm$) and azimuthal ($L_\pm$) operators:

$$|n,l,m\rangle \xrightarrow{T_+} |n{+}1,l,m\rangle \xrightarrow{L_+} |n{+}1,l,m{+}1\rangle \xrightarrow{T_-} |n,l,m{+}1\rangle \xrightarrow{L_-} |n,l,m\rangle. \tag{15}$$

The Berry phase for this plaquette is the argument of the holonomy product:

$$\theta = \arg\big(\langle n{+}1,l,m|T_+|n,l,m\rangle\,\langle n{+}1,l,m{+}1|L_+|n{+}1,l,m\rangle\,\langle n,l,m{+}1|T_-|n{+}1,l,m{+}1\rangle\,\langle n,l,m|L_-|n,l,m{+}1\rangle\big) \tag{16}$$

Each matrix element is a Biedenharn–Louck coupling coefficient. For the $SU(2)$ angular-momentum operators, $\langle l,m+1|L_+|l,m\rangle = \sqrt{(l-m)(l+m+1)}$, and for the $SU(1,1)$ radial operators, $\langle n+1,l|T_+|n,l\rangle = \sqrt{(n-l)(n+l+1)/4}$. **[SYMBOLIC PROOF]** All four factors are **real and positive**. Therefore

$$\theta = \arg(\text{positive real number}) = 0 \qquad \text{for every plaquette at every } (n,l,m). \tag{17}$$

The lattice connection defined by these operators is *flat*: it carries zero Berry curvature.

Erratum. An earlier version of this paper reported a fitted exponent $k = 2.113 \pm 0.015$ for the Berry phase scaling. That value was a theoretical prediction placed as a placeholder pending numerical validation; the accompanying figure was never generated (it remained a `[Placeholder]`). Subsequent computation confirms $\theta = 0$ identically for all plaquettes, as shown above. The claimed $k = 2.113$ is therefore **retracted**.

B. Log-Holonomy of Edge Weights

Although the Berry phase vanishes, the lattice edge weights carry a well-defined *log-holonomy*—the logarithmic circulation around a plaquette:

$$\Theta(n) = \ln w_{12} + \ln w_{23} - \ln w_{34} - \ln w_{41}, \tag{18}$$

where w_{ij} are the edge weights of the *topological-weight* lattice (`GeometricLattice(topological_weights=True)`); on the binary production lattice and on the CG-magnitude construction of §III the T-edge weight is m-independent and the L-edge weight is n-independent, so $\Theta \equiv 0$ there (2026-09-03). **[SYMBOLIC PROOF]** For the topological weights $w = 1/(n_1 n_2)$, the $T_\pm$ edges share the factor $1/\big(n(n+1)\big)$ and cancel, leaving

$$\Theta(n) = \ln\frac{1}{(n+1)^2} - \ln\frac{1}{n^2} = -2\ln\frac{n+1}{n} \;\sim\; -\frac{2}{n} \qquad (n \gg 1). \tag{19}$$

Both results—$\theta = 0$ on every plaquette (with a phased-plaquette control confirming the detector is not vacuous) and the exact $\Theta(n)$ closed form with its n^{-1} decay—are pinned by `tests/test_paper1_geometric_phase.py` (added 2026-09-01, closing a coverage gap: the prior Berry-phase artifact was archived with the retracted $k = 2.113$ claim). The per-plaquette log-holonomy $\Theta(n) = -2\ln((n+1)/n)$ decays as n^{-1} asymptotically (a closed form, so no fitted exponent or R^2 is involved). This measures the **curvature of the weight function**—how edge weights vary across the lattice—not a genuine geometric phase in the Berry sense.

C. Toward a Genuine Geometric Phase

A nonzero Berry phase requires complex-valued transition amplitudes. Possible extensions include:

1. the Condon–Shortley phase convention, which attaches $(-1)^m$ to $Y_{\ell m}$ and hence flips the sign of alternate $L_\pm$ matrix elements (convention-dependent, yielding discrete $\{0,\pi\}$ phases);
2. A $U(1)$ gauge extension—attaching phase factors $e^{if(n,l,m)}$ to the transition operators—which would produce continuously varying curvature;

3. An adiabatic Berry connection from eigenstates of a parameter-dependent Hamiltonian (as in the hyperspherical $P_{\mu\nu}(R)$ connection of Ref. [7]).

[OPEN] Whether any such extension recovers a scaling exponent near $k \approx 2$ (the velocity-dependent kinematic factor $v^2 \propto n^{-2}$) remains an open question for future work.

V. DISCUSSION: THE ALGEBRA-GEOMETRY DUALITY

Our results reveal a profound duality in the structure of the paraboloid lattice that mirrors the wave-particle duality of quantum mechanics.

A. The Dual Framework

1. **Algebraic Structure (Wave-like):** The operators $T_\pm, L_\pm$ and their eigenvalues reproduce the *unperturbed* hydrogen quantum-number structure exactly on interior shells (and the $SO(4)$ Casimir n^2-1 is exact on every shell); the energies (Eq. 5) are the imposed energy-shell map of n, not an output. This represents the continuous, exact aspect—analogous to the wave description in quantum mechanics.

2. **Geometric Structure (Particle-like):** The graph Laplacian $L = D - A$ carries finite-truncation structure that is intrinsic to its disconnected two-dimensional grid blocks; it supplies no physical correction or perturbation to the hydrogen levels. The weighted degree matrix D differs between angular momentum sectors at finite truncation (read before 2026-09-03 as differential connectivity [retracted 2026-09-04: sp-splitting-aliasing-mechanism] lifting the s/p degeneracy; withdrawn — the sectors are disconnected components). The edge-weight log-holonomy decays as n^{-1} (Section IV), measuring the curvature of the weight function. This represents the discrete aspect of the lattice formulation.

This duality is suggestive: the algebraic operators provide the exact quantum-number structure, while the graph topology introduces finite-size artifacts that must be controlled.

B. Why Calibration Fails

A critical observation: when we attempt to match the graph Laplacian's eigenvalues directly to physical energies, calibration fails. Eigenvalues remain non-negative (Section III), far from the negative binding energies of hydrogen.

The reason is structural: the Laplacian's spectrum is the band structure of its disconnected two-dimensional grid blocks, lying in $[0, 8]$, and it carries neither absolute energies nor physical splittings between hydrogen levels (Paper 7). The algebraic operators provide the quantum numbers, and the energies are the imposed energy-shell map of n; the graph Laplacian supplies no correction to them.

C. The Absence of Fine Structure

[MEASURED] A comprehensive computational search for the fine structure constant $\alpha \approx 1/137$ within the lattice geometry yields a null result. We examined:

1. **Operator ratios: [PANEL-VERIFIED]** The spectral-norm gearing ratio $\|L_+\|/\|T_+\| = 2.0$ *exactly*—at every lattice cutoff tested, not proved for all $n_{\max}$—while the Frobenius-norm ratio converges to $2\sqrt{6}/3 = 1.632993\ldots$, its exact limit (algebraic closed form; the frozen `test_trunk_qa_gearing.py` panel verifies only the monotone approach to $n_{\max} = 30$, ratio 1.701, not the limit) from $\|L_+\|_F^2/\|T_+\|_F^2 = 4(N+2)(2N+1)/(3N^2-2) \to 8/3$ (corrected 2026-09-03 from "drifts through $\approx$ 1.77 without converging": the sequence is monotone decreasing and 1.77 is a finite-$n_{\max}$ value, not a non-limit) (`tests/test_trunk_qa_gearing.py`). Neither encodes a physical constant such as 137 or $1/\alpha$: the spectral value is 2, and the drifting one is not a constant. The commutator $[T_+, L_+] = 0$, however, *is* bit-exact.

2. **Commutator defects: [PANEL-VERIFIED]** The operators commute exactly, $[T_+, L_+] = 0$, indicating perfect integrability rather than non-commutative geometry.

3. **Holonomy defects: [SYMBOLIC PROOF]** The Berry phase is identically zero for real operators (Section IV); the log-holonomy decays as n^{-1}, reflecting the asymptotic flatness of the weight function.

This null result is *physically correct*. The fine structure constant $\alpha = e^2/(4\pi\epsilon_0\hbar c)$ measures the strength of electromagnetic interactions—specifically, the coupling between electron and photon fields. Our lattice describes the *kinematic* structure of the electron state space without electromagnetic interactions. The hydrogen spectrum $E_n = -1/(2n^2)$ is independent of α; fine structure appears only through:

$$\text{Relativistic corrections:} \quad \Delta E_{\text{rel}} \sim \alpha^2 \cdot E_n/n, \tag{20}$$

$$\text{Spin-orbit coupling:} \quad \Delta E_{\text{so}} \sim \alpha^2 \cdot E_n \cdot l(l+1)/n^3, \tag{21}$$

$$\text{Lamb shift (QED):} \quad \Delta E_{\text{Lamb}} \sim \alpha^5 \cdot mc^2. \tag{22}$$

These require photon degrees of freedom, which are absent from our single-particle lattice.

This clarifies the role of geometry: the lattice encodes *kinematic* constraints (quantum-number labels, angular momentum structure) but not *dynamical* couplings. To incorporate fine structure, one must introduce a photon lattice and define interaction edges weighted by α. The fine structure constant would then appear as the *coupling strength between lattices*, not as an intrinsic geometric property of either lattice alone.

D. Artifact vs. Physics: Drawing the Line

Our convergence analysis establishes a clear separation between discretization artifacts and genuine physical content of the graph Laplacian. The s/p splitting is firmly in the artifact category: on the binary lattice it decays as $n_{\max} \to \infty$ (not on the CG-weighted construction of §III) and exhibits the oscillatory decay characteristic of finite-size aliasing [retracted 2026-09-04: sp-splitting-aliasing-mechanism] — a characterisation withdrawn for this quantity on 2026-09-03.

By contrast, the spectral bound $\lambda_{\max} \to 2d_{\max} = 8$ (hence the extremal eigenvalue $\kappa\lambda_{\max} \to -1/2$; the graph spectrum is *not* a level-by-level Rydberg ladder, `tests/test_paper7_graph_convergence.py`) and the edge-weight log-holonomy ($\Theta \propto n^{-1}$, Section IV) are *convergent* properties of the graph as $n_{\max} \to \infty$. They are structural content of the graph itself, not of S^3: the bound $8 = 2d_{\max}$ is the band edge of any large two-dimensional grid, and the graph Laplacian does not converge to Δ_{S^3} (Paper 7; the "survive the continuum limit" reading printed here until 2026-09-25 is withdrawn).

This distinction is essential: the graph Laplacian does not generate electromagnetic forces or the Lamb shift, and it does not carry the *kinematic* structure of the electron state space either. That structure lives in the label set (n, ℓ, m) and its ladder operators $T_\pm, L_\pm$ (with the $SO(4)$ Casimir $n^2 - 1$ diagonal on the labels); the graph Laplacian carries only the finite-truncation structure of its disconnected two-dimensional grid blocks. Finite lattice effects must be understood as numerical artifacts, not new physics.

E. The Wavefunction as Distribution

If quantum states are labelled by discrete nodes, what is the wavefunction $\psi(r)$? In this framework it is the standard object: a probability amplitude expanded in the basis states the nodes label, each node carrying one coefficient. A reading in which ψ is a statistical distribution over lattice sites and the Schrödinger equation becomes a diffusion equation on the graph is *not* available here: the graph Laplacian does not converge to Δ_{S^3} and its spectrum does not carry the Rydberg ladder (Paper 7; §III), so diffusion on the graph is not hydrogen dynamics (reading withdrawn 2026-09-28).

VI. CONCLUSION

We have demonstrated that the hydrogen state space, when labelled on a discrete paraboloid lattice, exhibits a dual structure:

- **Algebraic exactness: [PANEL-VERIFIED]** Transition operators reproduce the hydrogen quantum numbers bit-exactly on interior shells ($n < n_{\max}$) and $\ell(\ell+1)$ on every shell; the $SO(4)$ Casimir $n^2 - 1$ is exact on every shell; the energies are the imposed energy-shell map of n (Paper 7), not an output.
- **Understood discretization artifacts: [INTERNAL THEOREM]** The graph Laplacian's node-amplitude proxy for the s/p gap is closed form on $n_{\max} \equiv 0 \pmod 3$ and is set by that residue rather than by proximity to a continuum limit (2026-09-03), and is large at finite $n_{\max}$ (up to 37% at $n_{\max} = 5$ on the binary production lattice, 15.6% at 8, 1.7% at 10), and is not spectral aliasing [retracted 2026-09-04: sp-splitting-aliasing-mechanism] (withdrawn 2026-09-03: the l-blocks are disconnected and carry no degeneracy to alias) nor an emergent physical force. The proxy oscillates (4.1% at $n_{\max} = 20$) and reads 0.39% at $n_{\max} = 30$ on the $n_{\max} \equiv 0 \pmod 3$ branch (1.65%, 2.58% at 28, 29) — a proxy value, not a recovery of the exact Coulomb degeneracy [retracted 2026-09-04: sp-splitting-aliasing-mechanism], which belongs to the continuum operator; on the CG-weighted construction of §III the lift does *not* decay (129, 68, 84 and 139% at $n_{\max} = 8, 10, 15, 20$; `tests/test_paper1_block_spectrum.py`), so the artifact-decay claim is scoped to the binary lattice, and both figures carry the conditioning caveat of §III.
- **Geometric phase structure: [SYMBOLIC PROOF]** The Berry phase vanishes identically for the real-valued $SU(2)/SU(1,1)$ operators; the edge-weight log-holonomy decays as $\Theta \propto n^{-1}$, measuring the curvature of the weight function. Whether a $U(1)$ gauge extension can produce a genuine geometric phase with relativistic scaling remains an open question.
- **Kinematic completeness: [OBSERVATION]** The label set and its ladder operators encode the α-independent single-electron structure—the Coulomb quantum-number structure, angular momentum structure, and geometric phase effects—while cleanly separating boundary artifacts from physical content.

These results support a discrete label-set reformulation of the hydrogen state space in which the algebraic operators carry the exact quantum-number structure on interior shells (the energies being the imposed energy-shell map of n) and the graph Laplacian introduces controlled, quantitatively understood finite-size artifacts. The algebra-geometry duality is suggestive of a connection to wave-particle complementarity, though we do not claim it establishes one. Critically, the graph Laplacian does not replace electromagnetic interactions or generate the Lamb shift; the kinematic structure of the electron lives in the label set and its ladder operators, while the graph Laplacian carries only finite-truncation grid structure.

VII. LIMITATIONS AND NEXT STEPS: MISSING PHOTON DEGREES OF FREEDOM

What This Model Does Not Include:
The paraboloid lattice presented here is explicitly a *single-particle* model describing only the electron's quantum state manifold. It does not include:

1. **Photon degrees of freedom:** The electromagnetic field ($U(1)$ gauge symmetry) is absent.
2. **Electron-photon coupling:** There is no mechanism for radiative transitions or electromagnetic interactions.
3. **The fine structure constant α:** As a pure electron-only model, α cannot appear. The fine structure constant measures the strength of electron-photon coupling—it describes the relationship between *two* manifolds, not a property of either one alone.

What Must Be Added:
To incorporate electromagnetic interactions, one must construct a second geometric structure:

- **A photon gauge fiber:** The $U(1)$ electromagnetic gauge symmetry suggests a circular or helical fiber structure attached to each electron state (n, l, m).
- **Coupling rules:** Edges connecting the electron lattice to the photon fiber must be defined. The weight of these inter-manifold connections would encode the coupling strength.
- **Symplectic impedance:** If both manifolds carry action (units of $\hbar$), their ratio may define a dimensionless coupling constant.

[CONJECTURE] We conjecture that the fine structure constant α emerges as the *geometric impedance*—the ratio of action densities between the electron paraboloid and the photon fiber. **[OBSERVATION]** A companion paper [6] records the related observation that α satisfies the cubic $\alpha^3 - K\alpha + 1 = 0$ with $K = \pi(B + F - \Delta)$ on the Hopf fibration. It concludes that α behaves as a projection constant, not derivable from any single principle; the impedance picture above thus remains unexplored.

Outlook:
This framework invites generalization. If hydrogen is a paraboloid, what geometries describe molecules, nuclei, or quantum fields? The answer may lie in higher-dimensional lattices—label sets of larger symmetry groups.

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Appendix A: Mathematical Definitions

1. Weighted Degree Matrix

The weighted degree matrix D is a diagonal matrix where each diagonal element represents the total connection strength of a node:

$$D_{ii} = \sum_j |A_{ij}|, \tag{A1}$$

where A_{ij} is the adjacency matrix constructed from transition operators. For complex-valued operators, the absolute value ensures symmetric (undirected) edge weights. This definition explains why degree values are non-integer (Table I)—they represent weighted sums of transition matrix elements, not simple edge counts.

2. Berry Phase and Log-Holonomy

For a plaquette (Eq. 15), the Berry phase is $\theta = \arg(\prod_i \langle s_i|\hat{U}_i|s_{i+1}\rangle)$. As shown in Section IV, all four matrix elements are real and positive, so $\theta = 0$ identically.

The log-holonomy (Eq. 18) provides a nonzero measure of weight-function curvature. For topological weights $w = 1/(n_1 n_2)$, the exact result is $\Theta(n) = -2\ln\big((n+1)/n\big)$, decaying as n^{-1}.

Appendix B: Computational Details

1. Lattice Construction

The paraboloid lattice was constructed for $1 \leq n \leq 10$, yielding 385 quantum states $|n, l, m\rangle$ with $0 \leq l < n$ and $-l \leq m \leq l$. Transition operators $T_\pm$ and $L_\pm$ were built using Biedenharn-Louck coupling coefficients for $SU(2)$ and $SU(1,1)$ representations.

2. Graph Laplacian Diagonalization

The adjacency matrix A was computed as the sum of absolute values of all transition operators:

$$A = |T_+| + |T_-| + |L_+| + |L_-|. \tag{B1}$$

The graph Laplacian $L = D - A$ was stored in compressed sparse row (CSR) format. The weighted degree matrix D of the CG-magnitude construction ranges from 0.71 to 19.9 with mean degree 12.0 at $n_{\max} = 10$ (faithful rebuild, 2026-09-03; the figures printed earlier, 0.854–20.624 and 12.42, belong to the non-reproducing table of §III).

Eigenvalue computation used the shift-invert Lanczos method (`scipy.sparse.linalg.eigsh`) with $\sigma = -1.0$, computing the lowest 20 eigenvalues. Diagonalization required 0.15 seconds on a standard workstation; **[MEASURED]** the sparse-eigenvalue timing is guarded sub-quadratic and the adjacency nonzero count near-linear (measured exponent 1.07 over $n_{\max} = 5$–30) by `tests/test_ov_scaling_rigorous.py` (multi-point log–log fits; the frozen bounds are < 1.8 on timing and < 1.15 on the nonzero count).

State identification was performed by computing overlap probabilities $|\langle n, l, m|\psi_k\rangle|^2$ for each eigenvector $|\psi_k\rangle$, selecting the eigenvector with maximal overlap for each target state.

3. Log-Holonomy Calculation

Rectangular plaquettes (Eq. 15) were identified by searching for closed paths in (n, m) space at fixed l. The Berry phase $\theta = \arg(\cdot)$ is identically zero for all plaquettes because every $SU(2)/SU(1,1)$ matrix element is real and positive (Section IV). The log-holonomy $\Theta(n) = -2\ln\big((n+1)/n\big)$ was computed analytically and verified numerically; its closed form decays as n^{-1} asymptotically (no fitted exponent is involved—a finite-n log–log fit gives -0.99 on $n = 10$–320).

Appendix C: Supplementary Data

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[2] V. Fock, "Zur Theorie des Wasserstoffatoms," Z. Phys. **98**, 145 (1935).

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[5] F. R. K. Chung, *Spectral Graph Theory*, CBMS Regional Conference Series in Mathematics, Vol. 92 (American Mathematical Society, Providence, RI, 1997).

[6] J. Loutey, "The Fine Structure Constant as a Spectral Coincidence on the Hopf Fibration," (companion paper, 2026).

[7] J. Loutey, "The Hyperspherical Lattice: Two-Electron Atoms as Coupled Channel Graphs," (companion paper, 2026).

[8] J. Loutey, "The Dimensionless Vacuum: Recovering the Schrödinger Equation from a Scale-Invariant Label Lattice," GeoVac Paper 7 (2026).

[9] The weighted degree $D_{ii} = \sum_j |A_{ij}|$ represents the total connectivity strength of node i. For transition matrices with complex coefficients, we use absolute values to obtain undirected edge weights, explaining the non-integer values in Table I.

TABLE I. **Historical values — not reproduced by any current construction** (flagged 2026-09-03). These figures are retained to document the original calculation; the production binary lattice gives *integer* degrees — $D_{2s} = 2$ and $D_{2p} = 3$ exactly, every edge weight being 1 — while the CG construction gives 0.7071, 1.9319, 3.8284, 2.9568, 5.4096, 6.1237, of which only the $3d$ row here reproduces. (Corrected 2026-09-04: this caption assigned the CG values 1.9 and 3.8 to the binary lattice, inverting the separation it was written to enforce.) What survives is the *qualitative* ordering $D_{2p} > D_{2s}$, which both constructions share. Weighted node degrees for select quantum states. The non-integer values arise from weighted sums of transition matrix elements (see Appendix A).

State (n, l, m)	Weighted Degree D_{ii}	Type	Energy (Laplacian)
$(1, 0, 0)$	0.854	1s	(ground state)
$(2, 0, 0)$	0.854	2s	0.0202
$(2, 1, 0)$	3.416	2p	0.0238
$(3, 0, 0)$	0.854	3s	—
$(3, 1, 0)$	4.270	3p	—
$(3, 2, 0)$	6.124	3d	—